Alireza Daneshvar

Machine Learning Engineer | Data Scientist

Canada | +1 514-515-3695 | alireza.dg1998@gmail.com | linkedin.com/in/ali-reza-daneshvar-garmroodi
github.com/Alirezad126 | ml-portfolio-iota.vercel.app

Professional Summary

Machine learning engineer with a background in mechanical and building engineering. My work spans control and optimization, reinforcement learning, physics-informed machine learning, and language-model applications, applied to building energy systems, energy storage, oil and gas well integrity, and production software. I work between research and engineering, from developing methods at Johns Hopkins University and Concordia University to building and deploying ML systems in production.

Work Experience

Concordia University — **Montreal, Quebec, Canada**
Machine Learning Researcher — *Oct 2025 – Present*

- Build probabilistic and predictive models of leakage risk in oil and gas wells from noisy, incomplete regulatory records.
- Define validation protocols to test model results and identify gaps in the underlying data; presented at OMAE 2026.

Johns Hopkins University — **Remote, United States**
Machine Learning Researcher — *Feb 2025 – Present*

- Develop differentiable predictive control (DPC) methods that turn constrained optimization into control policies evaluated in a single forward pass.
- Design sparse, interpretable control policies with safe online adaptation, trained end to end through physics-based dynamics.
- Build simulation environments and benchmarks comparing optimization-based and learning-based controllers.
- Contribute differentiable optimization components to NeuroMANCER (PNNL) through pull requests, testing, and code review.

Concordia University — **Montreal, Quebec, Canada**
Research Assistant — *May 2021 – Apr 2023*

- Developed a safe deep reinforcement learning method, combining primal-dual optimization and imitation learning, for scheduling an energy hub with compressed air energy storage.
- Built regression surrogate models of nonlinear plant dynamics to keep long-horizon optimization tractable.
- Benchmarked against deep RL algorithms and MILP; published in the Journal of Energy Storage.

IranBar — **Tehran, Iran**
Data Scientist — *Jun 2019 – May 2020*

- Shipped machine learning models into a production pricing platform, cutting quotation time by 30% and raising agreement rates by 25%.
- Maintained deployed models with versioning, CI/CD, monitoring, and scheduled retraining.

Technical Skills

Machine Learning: Supervised Learning, Deep Learning, Reinforcement Learning, Optimization, Model Evaluation
NLP & LLMs: Retrieval-Augmented Generation, Agentic AI, Vector Databases, Fine-Tuning
Programming: Python, PyTorch, scikit-learn, pandas, NumPy, Git
Database and Deployment: PostgreSQL, NoSQL, Snowflake, Docker, FastAPI, CI/CD, MLflow, AWS

Education

Concordia University — **Montreal, Quebec, Canada**
Master of Applied Science (M.A.Sc.) in Building Engineering — *Jan 2021 – Apr 2023*

Iran University of Science and Technology — **Tehran, Iran**
Bachelor of Applied Science (B.A.Sc.) in Mechanical Engineering — *Sep 2016 – Aug 2020*

Publications

- A. Daneshvar Garmroodi and J. Drgona. **"SD-DPC: Sparse Dictionary Differentiable Predictive Control"** 2026 (Submitted). Link
- T. Banker, A. Daneshvar Garmroodi, N. Lawrence, J. Drgona, and A. Mesbah. **"Learning Safe Control Policies: Reinforcement Learning-Enabled Differentiable Predictive Control."** *IFAC World Congress*, 2026.
- A. Daneshvar Garmroodi, F. Nasiri, and F. Haghighat. **"Optimal Dispatch of an Energy Hub with Compressed Air Energy Storage: A Safe Reinforcement Learning Approach."** *Journal of Energy Storage*, 2022. Link

Relevant Projects

RAG-GPT: Document Question-Answering Chatbot **GitHub**

Python, LangChain, OpenAI API, FastAPI, React, Vector Database, Docker

- Built the parsing, chunking, embedding, and retrieval pipeline, served through a Dockerized FastAPI backend.

AI Chatbot With Retrieval-Augmented Generation

Python, LangChain, GPT-3.5-Turbo, Chroma, FastAPI, ReactJS, AWS EC2, Docker

- Deployed a public conversational assistant on AWS that grounds its answers in retrieved passages.

Fine-Tuning FLAN-T5 for Dialogue Summarization

Python, PyTorch, Hugging Face, PEFT, QLoRA, RLHF

- Fine-tuned with QLoRA, aligned with RLHF, and evaluated against the base and supervised-only models.

Building Control With Safe RL and Differentiable Predictive Control **GitHub**

Python, PyTorch, PD-DDPG, OpenAI Gym, NeuroMANCER

- Benchmarked primal-dual DDPG against DPC in a physics-based Gym environment, released as a worked example in NeuroMANCER.